MATH 323: Probability

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Abstract

Contents

1 Introduction

These notes were written for MATH 323 (Probability) at McGill University, Summer 2026. The course is taught by Magid Sabbagh and meets in Stewart Biology Building S1/3 every Monday, Tuesday, Wednesday, and Thursday from 8:35 am to 10:55 am.

Grading scheme

The final grade is the higher of the two:

$$20\% \text{ assignments} + 5\% \text{ multiple choice quiz} + 25\% \text{ midterm} + 50\% \text{ final} \tag{1}$$
$$\text{or} \tag{2}$$
$$20\% \text{ assignments} + 5\% \text{ multiple choice quiz} + 75\% \text{ final} \tag{3}$$

The textbook is *Mathematical Statistics with Applications*, 7th edition. The midterm is 1h30 long but students get the full 2h30 class block to complete it.

2 Prerequisite Knowledge

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2.1 Sets and Set Operations

A **set** is a well-defined collection of objects, where "well-defined" means we can determine whether or not a given element x belongs to the set. We write $a \in A$ if a is an element of A, and $a \notin A$ otherwise. A set is specified either by listing its elements or by giving a property that defines membership.

We always work inside a fixed **universal set** Ω. Let $A, B \subseteq \Omega$. We define the following operations:

- **Subset**: $A \subseteq B$ if $x \in A$ implies $x \in B$.
- **Equality**: $A = B$ if $A \subseteq B$ and $B \subseteq A$.
- **Proper subset**: $A \subsetneq B$ if $A \subseteq B$ and $A \neq B$.
- **Union**: $A \cup B = \{x : x \in A \text{ or } x \in B\}$.
- **Intersection**: $A \cap B = \{x : x \in A \text{ and } x \in B\}$.
- **Difference**: $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$.
- **Complement**: $A^c = \{x : x \in \Omega \text{ and } x \notin A\}$, also written $\overline{A}$.
- **Singleton**: $\{x\} = \{y : y = x\}$.

The empty set is denoted $\emptyset$.

Proposition 1. *For any $A \subseteq \Omega$:*

$$A \cup A^c = \Omega, \qquad A \cap A^c = \emptyset, \qquad A \cup \emptyset = A, \qquad A \cap \emptyset = \emptyset.$$

Proposition 2. *Let $A, B, C \subseteq \Omega$. Then:*

1. ***Commutativity:*** *$A \cup B = B \cup A$ and $A \cap B = B \cap A$.*

2. ***Associativity:*** *$A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$.*

3. ***Distributivity:*** *$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.*

Proof. We prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$; the other identities are similar.

($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$, so $x \in B$ or $x \in C$. If $x \in B$, then $x \in A \cap B$. If $x \in C$, then $x \in A \cap C$. In either case $x \in (A \cap B) \cup (A \cap C)$.

($\supseteq$) Let $x \in (A \cap B) \cup (A \cap C)$. If $x \in A \cap B$, then $x \in A$ and $x \in B \subseteq B \cup C$, so $x \in A \cap (B \cup C)$. The case $x \in A \cap C$ is analogous. $\square$

Proposition 3 (De Morgan's Laws)**.** *For $A, B \subseteq \Omega$:*

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

2.2 Indexed Unions and Intersections

Let $A_1, A_2, A_3, \ldots$ be subsets of Ω. For $n \in \mathbb{N}$:

$$\bigcap_{i=1}^{n} A_i = \{x \in \Omega : x \in A_i \text{ for all } i \in \{1, \ldots, n\}\},$$
$$\bigcup_{i=1}^{n} A_i = \{x \in \Omega : x \in A_i \text{ for at least one } i \in \{1, \ldots, n\}\}.$$

The countable versions are defined the same way with i ranging over $\mathbb{N}$:

$$\bigcap_{i=1}^{\infty} A_i = \{x \in \Omega : x \in A_i \text{ for all } i \in \mathbb{N}\}, \qquad \bigcup_{i=1}^{\infty} A_i = \{x \in \Omega : x \in A_i \text{ for some } i \in \mathbb{N}\}.$$

Proposition 4 (Generalized De Morgan)**.**

$$\left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c, \qquad \left(\bigcap_{i=1}^{\infty} A_i\right)^c = \bigcup_{i=1}^{\infty} A_i^c.$$

2.3 Cardinality and Inclusion-Exclusion

Problem 1. *In a class of 101 students, 20 play hockey, 30 like math, and 5 do both. How many students:*

1. *do not play hockey?*

2. *do not like math?*

3. *neither play hockey nor like math?*

4. *play hockey but do not like math?*

Solution 1. *1.* $101 - 20 = 81$.

2. $101 - 30 = 71$.

3. $101 - (20 + 30 - 5) = 56$.

4. $20 - 5 = 15$.

We have the identity $A \setminus B = A \cap B^c = A \setminus (A \cap B)$.

Remark 1. *Write $|A|$ for the cardinality of A. Then*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Proof. Partition $A \cup B$ into three disjoint pieces: $A \setminus B$, $A \cap B$, and $B \setminus A$. Let $x = |A \setminus B|$, $y = |A \cap B|$, and $z = |B \setminus A|$. Then $|A \cup B| = x + y + z$, $|A| = x + y$, and $|B| = y + z$, so

$$|A| + |B| - |A \cap B| = (x + y) + (y + z) - y = x + y + z = |A \cup B|. \qquad \square$$

3 Sigma-Algebras and Probability Measures

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3.1 Sigma-Algebras

Definition 1. *A **σ-algebra** on a set Ω is a collection $\mathbb{F}$ of subsets of Ω such that:*

1. $\emptyset \in \mathbb{F}$,

2. *if $A \in \mathbb{F}$, then $A^c \in \mathbb{F}$,*

3. *if $A_1, A_2, \ldots \in \mathbb{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathbb{F}$.*

*Elements of $\mathbb{F}$ are called **events**.*

Example.

1. *Let Ω be any set. Then $\mathbb{F} = \{\emptyset, \Omega\}$ is a σ-algebra (the trivial one).*

2. *Let Ω be any set. The power set 2^{Ω} (the collection of all subsets of Ω) is a σ-algebra.*

3. *Let Ω be a set. Then $\mathbb{F} = \{A \subseteq \Omega : A$ is countable or A^c is countable$\}$ is a σ-algebra.*

4. *For any $A \subseteq \Omega$, the collection $\mathbb{F} = \{\emptyset, A, A^c, \Omega\}$ is a σ-algebra.*

Example. *Flip a coin once. Then $\Omega = \{H, T\}$ and $2^{\Omega} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$.*

3.2 Probability Measures

Definition 2. *Let Ω be the sample space of an experiment and $\mathbb{F}$ a σ-algebra of subsets of Ω. A* ***probability measure*** *is a function $P : \mathbb{F} \to [0, 1]$ satisfying the* ***Kolmogorov axioms****:*

1. *$P(A) \geq 0$ for all $A \in \mathbb{F}$.*
2. *$P(\Omega) = 1$.*
3. ***Countable additivity****: if $A_1, A_2, \ldots \in \mathbb{F}$ are pairwise disjoint (that is, $A_k \cap A_l = \emptyset$ whenever $k \neq l$), then*

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

The triple $(\Omega, \mathbb{F}, P)$ is called a ***probability space****.*

3.3 Basic Properties of Probability

Proposition 5 (Monotonicity)**.** *If $A \subseteq B$, then $P(A) \leq P(B)$.*

Proof. Write $B = A \cup (B \setminus A)$ as a disjoint union. By countable additivity, $P(B) = P(A) + P(B \setminus A) \geq P(A)$, since $P(B \setminus A) \geq 0$. $\square$

Corollary 0.1. *For any event A, $P(A) \leq 1$.*

Proof. Since $A \subseteq \Omega$, monotonicity gives $P(A) \leq P(\Omega) = 1$. $\square$

Proposition 6 (Complement Rule)**.** *$P(A^c) = 1 - P(A)$.*

Proof. Since $\Omega = A \cup A^c$ disjointly, $1 = P(\Omega) = P(A) + P(A^c)$, so $P(A^c) = 1 - P(A)$. $\square$

Proposition 7 (Inclusion-Exclusion for Two Events)**.** *For any events $A, B \in \mathbb{F}$:*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof. Decompose $A \cup B = A \cup (B \cap A^c)$ where A and $B \cap A^c$ are disjoint. By countable additivity,

$$P(A \cup B) = P(A) + P(B \cap A^c). \tag{$*$}$$

Similarly, $B = (A \cap B) \cup (A^c \cap B)$ is a disjoint decomposition, so

$$P(B) = P(A \cap B) + P(A^c \cap B), \qquad \text{i.e.} \qquad P(A^c \cap B) = P(B) - P(A \cap B).$$

Substituting this into $(*)$ gives $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. $\square$

3.4 Boole's Inequality and Bonferroni

The countable additivity axiom requires the events to be pairwise disjoint. What can we say about $P(\bigcup_{i=1}^{\infty} A_i)$ when the A_i are not disjoint?

Proposition 8 (Boole's Inequality / Union Bound)**.** *For any sequence of events* $\{A_i\}_{i=1}^{\infty}$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i).$$

Proof. Define a new sequence by "carving out" the previously seen sets:

$$A_1^\star = A_1, \qquad A_n^\star = A_n \setminus \bigcup_{i=1}^{n-1} A_i \quad \text{for } n \geq 2.$$

Step 1: The $A_i^\star$ are pairwise disjoint. Suppose for contradiction that $A_k^\star \cap A_l^\star \neq \emptyset$ with $k < l$. Pick x in the intersection. Since $x \in A_l^\star$, by definition $x \notin \bigcup_{i=1}^{l-1} A_i$, in particular $x \notin A_k$. But $x \in A_k^\star \subseteq A_k$, contradiction.

Step 2: $\bigcup_{i=1}^{\infty} A_i^\star = \bigcup_{i=1}^{\infty} A_i$. Since $A_i^\star \subseteq A_i$ for all i, the inclusion $\bigcup A_i^\star \subseteq \bigcup A_i$ is immediate. For the reverse, let $x \in \bigcup A_i$ and let i_0 be the smallest index with $x \in A_{i_0}$. Then $x \notin A_i$ for $i < i_0$, so $x \in A_{i_0}^\star \subseteq \bigcup A_i^\star$.

Step 3: Conclude. By countable additivity applied to the $A_i^\star$ together with monotonicity ($A_i^\star \subseteq A_i$ implies $P(A_i^\star) \leq P(A_i)$),

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\bigcup_{i=1}^{\infty} A_i^\star\right) = \sum_{i=1}^{\infty} P(A_i^\star) \leq \sum_{i=1}^{\infty} P(A_i). \qquad \square$$

Corollary 0.2 (Bonferroni Inequality)**.** *For events* $A_1, \ldots, A_n$,

$$P\left(\bigcap_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} P(A_i) - (n-1).$$

Proof. Using De Morgan's law, the complement rule, and the union bound,

$$\begin{aligned} P\left(\bigcap_{i=1}^{n} A_i\right) &= 1 - P\left(\bigcup_{i=1}^{n} A_i^c\right) \\ &\geq 1 - \sum_{i=1}^{n} P(A_i^c) \\ &= 1 - \sum_{i=1}^{n} \big(1 - P(A_i)\big) \\ &= \sum_{i=1}^{n} P(A_i) - (n-1). \square \end{aligned}$$

Problem 2. *Suppose* $\{A_i\}_{i=1}^{\infty}$ *are events with* $P(A_i) = 1$ *for all* $i \in \mathbb{N}$*. Show that* $P(\bigcap_{i=1}^{\infty} A_i) = 1$.[1]

[1] He'll prolly ask this on an exam.

3.5 Independent Events

Definition 3. *Two events A and B are **independent** if $P(A \cap B) = P(A) \cdot P(B)$.*

4 Finite Sample Spaces and Equally Likely Outcomes

Example. *Roll a die twice, with the rolls independent, and record the pair of outcomes. The sample space is*

$$\Omega = \{(a, b) : a, b \in \{1, 2, 3, 4, 5, 6\}\}, \qquad |\Omega| = 36.$$

By independence, $P(\{(a, b)\}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$, so all 36 outcomes are equally likely.

Definition 4. *Let $\Omega = \{w_1, \ldots, w_N\}$ be a finite sample space with $|\Omega| = N$. The outcomes are **equally likely** if*

$$P(\{w_1\}) = P(\{w_2\}) = \cdots = P(\{w_N\}) = \tfrac{1}{N}.$$

For any event $A \subseteq \Omega$ with $|A| = d$, writing $A = \{w_{i_1}, \ldots, w_{i_d}\}$,

$$P(A) = \sum_{k=1}^{d} P(\{w_{i_k}\}) = \frac{d}{N} = \frac{|A|}{|\Omega|}.$$

Example. *A fair coin is flipped twice independently. The sample space is $\Omega = \{HH, HT, TH, TT\}$ with each outcome having probability $\frac{1}{4}$.*

What is the probability of getting at least one head? Let A be that event. Then $A = \{HH, HT, TH\}$, so $|A| = 3$ and $P(A) = \frac{3}{4}$.

5 Counting

Theorem 1 (Fundamental Counting Principle)**.** *Given m elements $a_1, \ldots, a_m$ and n elements $b_1, \ldots, b_n$, the number of distinct ordered pairs (a, b) with $a \in \{a_1, \ldots, a_m\}$ and $b \in \{b_1, \ldots, b_n\}$ is mn.*

Example. *How many 7-digit telephone numbers can be formed?*

Each digit has 10 choices, giving 10^7 numbers.

How many if the third digit must be even? The third digit has 5 choices ($0, 2, 4, 6, 8$) and each of the other six digits has 10 choices, giving $10^6 \cdot 5$.

5.1 Permutations

Example. *In a class of 100 students, 2 are chosen: the first wins \$100 and the second wins \$75. How many choices are there if:*

1. *a student can win twice (**sampling with replacement**)?*

2. *a student can win at most once (**sampling without replacement**)?*

Solution 2. 1. *100 choices for the first prize and 100 for the second:* $100^2 = 10{,}000$.

2. *100 choices for the first prize and 99 for the second:* $100 \cdot 99 = 9{,}900$.

Definition 5. *An ordered arrangement of r distinct objects chosen from n objects (without replacement) is called a **permutation**. The number of such permutations is*

$$P_r^n = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

Example. *In a class of 100 students, 2 students are chosen to form a committee where order does not matter. How many committees are possible?*

There are $100 \cdot 99$ ordered pairs, but each unordered committee is counted $2!$ times, so the number of committees is

$$\frac{100 \cdot 99}{2!} = 4{,}950.$$

6 Appendix

7 Solutions

8 Useful Links